

## *Scanning the human proteome for calmodulin-binding proteins*

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The calcium ion ( $\text{Ca}^{2+}$ ) is a ubiquitous second messenger that is crucial for the regulation of a wide variety of cellular processes. The diverse transient signals transduced by  $\text{Ca}^{2+}$  are mediated by intracellular  $\text{Ca}^{2+}$ -binding proteins, also known as  $\text{Ca}^{2+}$  sensors. A key obstacle to studying many  $\text{Ca}^{2+}$ -sensing proteins is the difficulty in identifying the numerous downstream target interactions that respond to  $\text{Ca}^{2+}$ -induced conformational changes. Among a number of  $\text{Ca}^{2+}$  sensors in the eukaryotic cell, calmodulin (CaM) is the most widespread and the best studied. Employing the mRNA display technique, we have scanned the human proteome for CaM-binding proteins and have identified and characterized a large number of both known and previously uncharacterized proteins that interact with CaM in a  $\text{Ca}^{2+}$ -dependent manner. The interactions of several identified proteins with  $\text{Ca}^{2+}$ /CaM were confirmed by using pull-down assays and coimmunoprecipitation. Many of the CaM-binding proteins identified belong to protein families such as the DEAD/H box proteins, ribosomal proteins, proteasome 26S subunits, and deubiquitinating enzymes, suggesting the possible involvement of  $\text{Ca}^{2+}$ /CaM in different signaling pathways. The selection method described herein could be used to identify the binding partners of other calcium sensors on the proteome-wide scale.